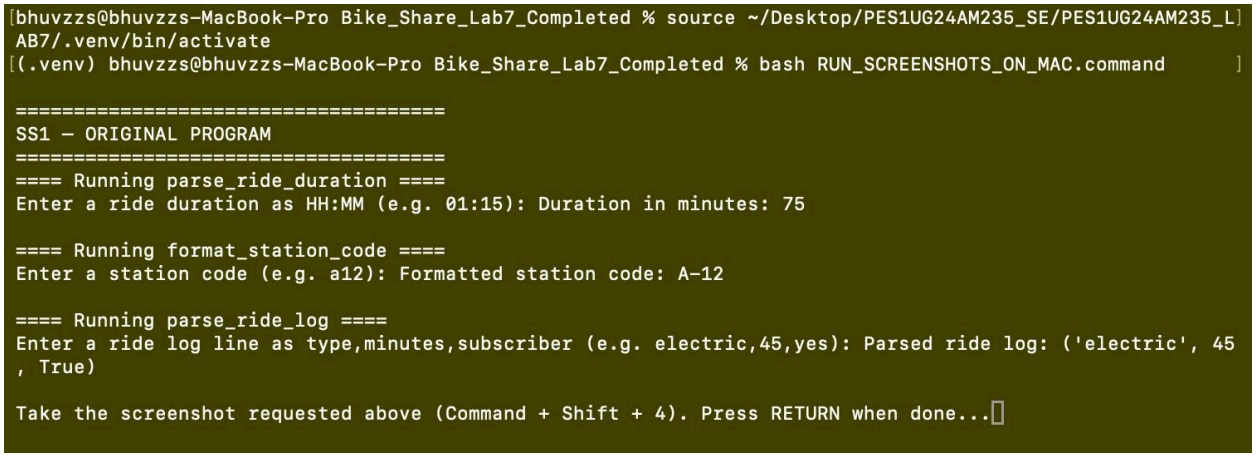
LAB 7: Fuzz Testing for Input Validation Bugs

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Application: Bike Share Rental

Screenshot 1 — Execution of Original Buggy Application (SS1)



Fuzz Testing Before Bug Fixes (SS2)

```
SS2 — FAILING FUZZ TESTS (EXPECTED)
====================================
============================= test session starts ==============================
platform darwin -- Python 3.13.15, pytest-9.2.0, pluggy-1.7.0 -- /Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.
venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE
plugins: hypothesis-6.170.15
collecting ... collected 3 items

test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash FAILED [ 33%]
test_bike_share_rental_parser.py::test_format_station_code_no_crash FAILED [ 66%]
test_bike_share_rental_parser.py::test_parse_ride_log_no_crash FAILED    [100%]

=================================== FAILURES ===================================
  + Exception Group Traceback (most recent call last):
  |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line 1
2, in test_parse_ride_duration_no_crash
  |     @example("")
  |
  |   File "/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py"
, line 1730, in _raise_to_user
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 18, in test_parse_ride_duration_no_crash
    |     result = parse_ride_duration(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 6, i
n parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 20, in test_parse_ride_duration_no_crash
    |     raise AssertionError(f"parse_ride_duration crashed for {s!r}") from exc
    | AssertionError: parse_ride_duration crashed for None
    | Failing explicit example: test_parse_ride_duration_no_crash(
    |     s=None,
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 18, in test_parse_ride_duration_no_crash
    |     result = parse_ride_duration(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 6, i
n parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | The above exception was the direct cause of the following exception:
    |
```

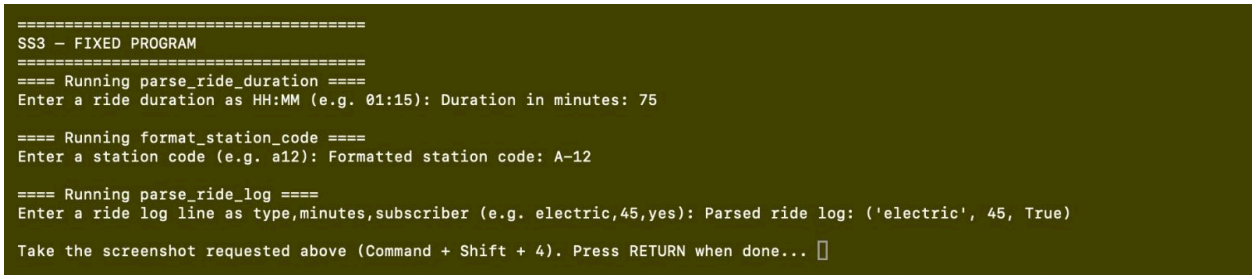
```
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 20, in test_parse_ride_duration_no_crash
    |     raise AssertionError(f"parse_ride_duration crashed for {s!r}") from exc
    | AssertionError: parse_ride_duration crashed for ''
    | Failing explicit example: test_parse_ride_duration_no_crash(
    |     s='',
    | )
    +------------------------------------
/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py:1730: Except
ionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
  + Exception Group Traceback (most recent call last):
  |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line 2
8, in test_format_station_code_no_crash
  |     @example("")
  |
  |   File "/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py"
, line 1730, in _raise_to_user
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 34, in test_format_station_code_no_crash
    |     result = format_station_code(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 14,
in format_station_code
    |     code = code.strip().upper()
    |            ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 36, in test_format_station_code_no_crash
    |     raise AssertionError(f"format_station_code crashed for {s!r}") from exc
    | AssertionError: format_station_code crashed for None
    | Failing explicit example: test_format_station_code_no_crash(
    |     s=None,
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 34, in test_format_station_code_no_crash
    |     result = format_station_code(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 15,
in format_station_code
    |     return code[0] + '-' + code[1:]
    |            ~~~~^^^
    | IndexError: string index out of range
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 36, in test_format_station_code_no_crash
    |     raise AssertionError(f"format_station_code crashed for {s!r}") from exc
    | AssertionError: format_station_code crashed for ''
```

```
 36, in test_format_station_code_no_crash
    |     raise AssertionError(f"format_station_code crashed for {s!r}") from exc
    | AssertionError: format_station_code crashed for ''
    | Failing explicit example: test_format_station_code_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 39, in test_format_station_code_no_crash
    |     assert result == "A-12"
    | AssertionError: assert 'A--12' == 'A-12'
    |
    |   - A-12
    |   + A--12
    |   ?  +
    | Failing explicit example: test_format_station_code_no_crash(
    |     s=' A-12 ',
    | )
    +------------------------------------
/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py:1730: Except
ionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
  + Exception Group Traceback (most recent call last):
  |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line 4
3, in test_parse_ride_log_no_crash
  |     @example("")
  |
  |   File "/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py"
, line 1730, in _raise_to_user
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 49, in test_parse_ride_log_no_crash
    |     result = parse_ride_log(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 22,
in parse_ride_log
    |     parts = line.split(',')
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 51, in test_parse_ride_log_no_crash
    |     raise AssertionError(f"parse_ride_log crashed for {s!r}") from exc
    | AssertionError: parse_ride_log crashed for None
    | Failing explicit example: test_parse_ride_log_no_crash(
    |     s=None,
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 49, in test_parse_ride_log_no_crash
    |     result = parse_ride_log(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 23,
in parse_ride_log
    |     return parts[0].strip(), int(parts[1]), parts[2].strip() == 'yes'
```

```
    |     return parts[0].strip(), int(parts[1]), parts[2].strip() == 'yes'
    |                              ~~~~~^^^
    | IndexError: list index out of range
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 51, in test_parse_ride_log_no_crash
    |     raise AssertionError(f"parse_ride_log crashed for {s!r}") from exc
    | AssertionError: parse_ride_log crashed for ''
    | Failing explicit example: test_parse_ride_log_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 49, in test_parse_ride_log_no_crash
    |     result = parse_ride_log(s)
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/bike_share_rental_parser.py", line 23,
in parse_ride_log
    |     return parts[0].strip(), int(parts[1]), parts[2].strip() == 'yes'
    |                              ~~~^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: 'not_a_number'
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    |   File "/Users/bhuvzzs/Downloads/Bike_Share_Lab7_Completed/BEFORE_FIX_REFERENCE/test_bike_share_rental_parser.py", line
 51, in test_parse_ride_log_no_crash
    |     raise AssertionError(f"parse_ride_log crashed for {s!r}") from exc
    | AssertionError: parse_ride_log crashed for 'electric,not_a_number,yes'
    | Failing explicit example: test_parse_ride_log_no_crash(
    |     s='electric,not_a_number,yes',
    | )
    +------------------------------------
/Users/bhuvzzs/Desktop/PES1UG24AM235_SE/PES1UG24AM235_LAB7/.venv/lib/python3.13/site-packages/hypothesis/core.py:1730: Except
ionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
=========================== short test summary info ============================
FAILED test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash - ...
FAILED test_bike_share_rental_parser.py::test_format_station_code_no_crash - ...
FAILED test_bike_share_rental_parser.py::test_parse_ride_log_no_crash - Excep...
============================== 3 failed in 0.84s ===============================

Take the screenshot requested above (Command + Shift + 4). Press RETURN when done...
```

Execution of Fixed Application (SS3)



Fuzz Testing After Bug Fixes (SS4)

